Solution Proposal by

<TRINETRA TECH>

The Third Eye That Watches, Tracks, and Protects Every Worker on Your Site

MEET THE TEAM

Ronak Kumar

(Team Lead)

"I don't just run the team.
I run the whole system."

Adithya Prajapati

(Attendance & PPE Detection)

"No PPE? My system knew
before you even walked in."

Manavi S

(Website & Dashboard)

"One screen to rule them
all. I built that screen."

Shruti Saumya

(Presentation & Site Work)

"I tell the story. The team
builds it. Fair deal."

PROBLEM STATEMENT

Problem Statement

Manual PPE compliance checks and health monitoring in construction sites cause critical safety gaps and delayed emergency response.

Problem Statement Expansion

Construction sites rely on manual inspections and reactive safety protocols, creating significant compliance blind spots, missed medical emergencies, and undetected environmental hazards. By 2026, the industry is rapidly shifting toward AI-enabled, real-time safety systems that integrate wearable health monitoring and environmental sensing to proactively protect workers and reduce incident rates.

How it manifests for the user or client

Workers enter hazardous zones without proper PPE undetected. Supervisors lack real-time visibility into workforce health and site conditions. Safety officers have no predictive data to act before incidents occur — relying entirely on post-incident reports and manual walkthroughs that are too slow, too infrequent, and too limited in scope.

How it impacts the user or client

In 2026, PPE non-compliance and undetected health emergencies lead to fatal injuries, permanent disabilities, and massive legal and financial liability for construction firms. For workers, the absence of real-time monitoring means a medical episode like heat stroke or cardiac arrest goes unnoticed until it's too late. For companies, regulatory fines, project shutdowns, and reputational damage create a high-cost cycle of reactive crisis management that could be entirely prevented.

USER PERSONA 1

RAJESH

- 32 years old
- Bangalore
- Construction worker
- ₹15,000-₹25,000/month

ABOUT THE USER

Rajesh works 10-hour shifts at high-rise construction sites, operating heavy machinery and working at dangerous heights. He supports his family and prioritizes workplace safety.

ABOUT THE USER

- Hardworking and dedicated
- Family-oriented provider
- Risk-aware but time-pressured

PROBLEMS

- Extreme heat causes fatigue and dizziness
- No real-time health monitoring
- Delayed emergency response

GOALS AND NEEDS

- Immediate help during emergencies
- Return home safely every day
- Protection from workplace hazards

CHALLENGES

- Inconsistent PPE usage compliance
- Language barriers with safety instructions
- Limited access to immediate medical care

USER PERSONA 2

PRIYA

- 38 years old
- Bangalore
- Site Supervisor
- ₹45,000 - ₹75,000/month

ABOUT THE USER

Priya manages 50+ workers across multiple construction zones. She is responsible for safety compliance, emergency coordination, and ensuring PPE usage while maintaining project timelines

PROBLEMS

- Cannot monitor all workers simultaneously
- Manual PPE verification is time-consuming
- Late detection of health emergencies

CHALLENGES

- Balancing safety and productivity demands
- Managing diverse workforce across zones
- Maintaining accurate documentation

ABOUT THE USER

- Detail-oriented and responsible
- Strong leadership and communication
- Safety-first mindset

GOALS AND NEEDS

- Real-time oversight of worker safety
- Automated compliance tracking
- Instant emergency notifications

Solution Details

Trinetra Tech

Trinetra is a three-part AI-powered occupational health and safety system for construction sites. The first part is an AI-based smart attendance system that marks a worker present only when full PPE compliance is detected - ensuring safety before entry. The second part is an autonomous rover that continuously patrols the site, monitoring workers and environmental conditions in real time. The third part is a centralized web dashboard that instantly alerts authorities the moment any single worker is identified as at risk - enabling zero-delay emergency response.

Demographic it Impacts

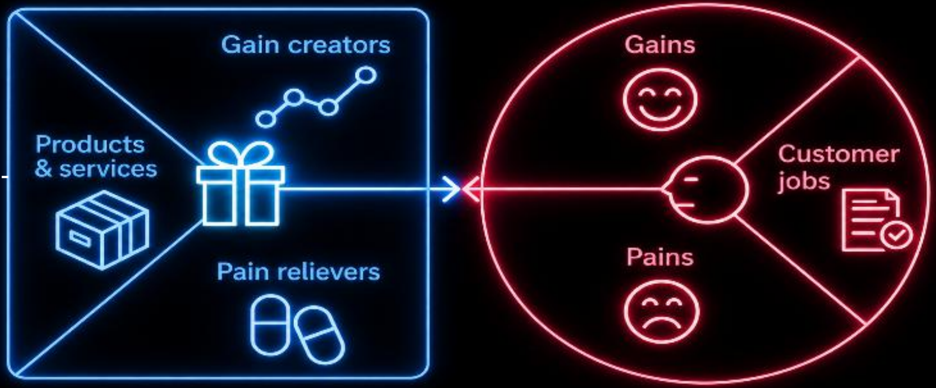
Construction workers (ensured PPE compliance & site safety), site supervisors (real-time rover-based workforce monitoring), and safety officers & authorities (instant alerts and centralized risk management).

Value proposition

These are AI-driven features — PPE compliance checks, real-time health monitoring, and risk scoring — that help supervisors ensure worker safety and exceed site safety standards

These are the benefits expected — zero PPE violations, early fatigue detection, real-time hazard alerts, and a centralized safety dashboard for remote supervision.

These are the actual offerings: ESP32 wearables, environmental sensors, ESP32-CAM PPE detection, Firebase cloud dashboard, and an AI risk engine



These are the tasks site supervisors and safety officers are trying to accomplish — monitoring worker health, enforcing PPE compliance, and preventing occupational hazards.

These address site safety challenges — automated PPE entry denial, fall detection alerts, gas/dust hazard warnings, and AI-predicted risk scores prevent accidents before they occur

These are the negative outcomes sites face — unreported health emergencies, PPE non-compliance, toxic air exposure, and delayed supervisor response to on-site dangers.

Thinking BIG

Current State vs Future State of our solution

CURRENT STATE - MVP

Data movement between components

→ **SENSORS & DEVICES** — Wearables, Environmental Sensors, ESP32
→ **ESP32 GATEWAY** — Collects data and transmits
→ **FIREBASE DATABASE** — Stores raw sensor data
→ **BASIC DASHBOARD** — Displays basic metrics & alerts
→ **SUPERVISOR / SAFETY TEAM** — Reviews data & takes action

LIMITATIONS

- Limited sensor integration
- No predictive analytics
- Manual alerting
- No AI risk intelligence
- Limited real-time visualization

NORTH STAR STATE

Future data movement – Integrated, inclusive & real-time

→ **ALL DEVICES & SENSORS** — Wearables, Environmental, PPE Camera, RFID, ESP32
→ **REAL-TIME DATA INGESTION** — Events captured in real-time from all sources
→ **CENTRALIZED CLOUD PLATFORM** — Unified, clean, secure & scalable (Firebase)
→ **AI ANALYTICS & RISK ENGINE** — Segmentation, predictions, risk scoring
→ **SMART DECISIONS & ACTIONS** — Automated alerts, insights & impact analysis

ENABLERS

- Complete data coverage
- Integrated & real-time data
- AI-driven risk intelligence
- Role-based dashboards
- Automated alerts & faster response

OUR VISION: AI-powered, real-time occupational safety ecosystem ensuring zero-incident construction sites.

Architecture Discovery

HopeWorks

AI4India

WORKFLOW & SYSTEM ASSUMPTIONS

Technical Assumptions

Customer Assumptions

- Architecture must be scalable, reliable, and support full system workload.
- Required development technologies, tools (ESP32, Firebase, specific sensors), and licenses are accessible.
- Continuous sensor data from 18+ countries will be available and processed in real-time.

Data Assumptions

- Development, test, and production environments for multi-country rollout are defined.
- Third-party data APIs for regional data are stable.

CRITICAL DESIGN & MODULE DECISIONS

Technical Decisions

- Choose Cloud Platform for Dashboard (e.g., Firebase, GCP, Azure).
- Adopt Microservices to separate Module 1 (PPE), Module 2 (Health), and Module 3 (Env).

Customer Decisions

- Select tech stack (e.g., ESP32-CAM, MAX30102, DS18B20, MPU6050, DHT22, MQ135).
- Define data ingestion strategy for diverse sensor inputs.

Data Decisions

- Decide on PPE AI model type and deployment (Cloud vs. Edge).

Team Alignment

Unified Vision for Safety-by-Design and Cross-Module Integration

MODULE-SPECIFIC RISKS & MITIGATION

Technical Risks · Customer Risks · Data Risks

Risk	Mitigation	Module
PPE Detection False Negatives	Refine AI model training, diverse data sources	Module 1
Sensor Failures (Wearables)	Sensor redundancy, robust connectivity checks	Module 2
Database Inconsistency	Firebase Real-time DB rules, ACID transactions	Database
Team Coordination Gaps	Inter-team APIs, regular syncs	Teams 1-5

DEVELOPMENT & SYSTEM DEPENDENCIES

Technical Dependencies

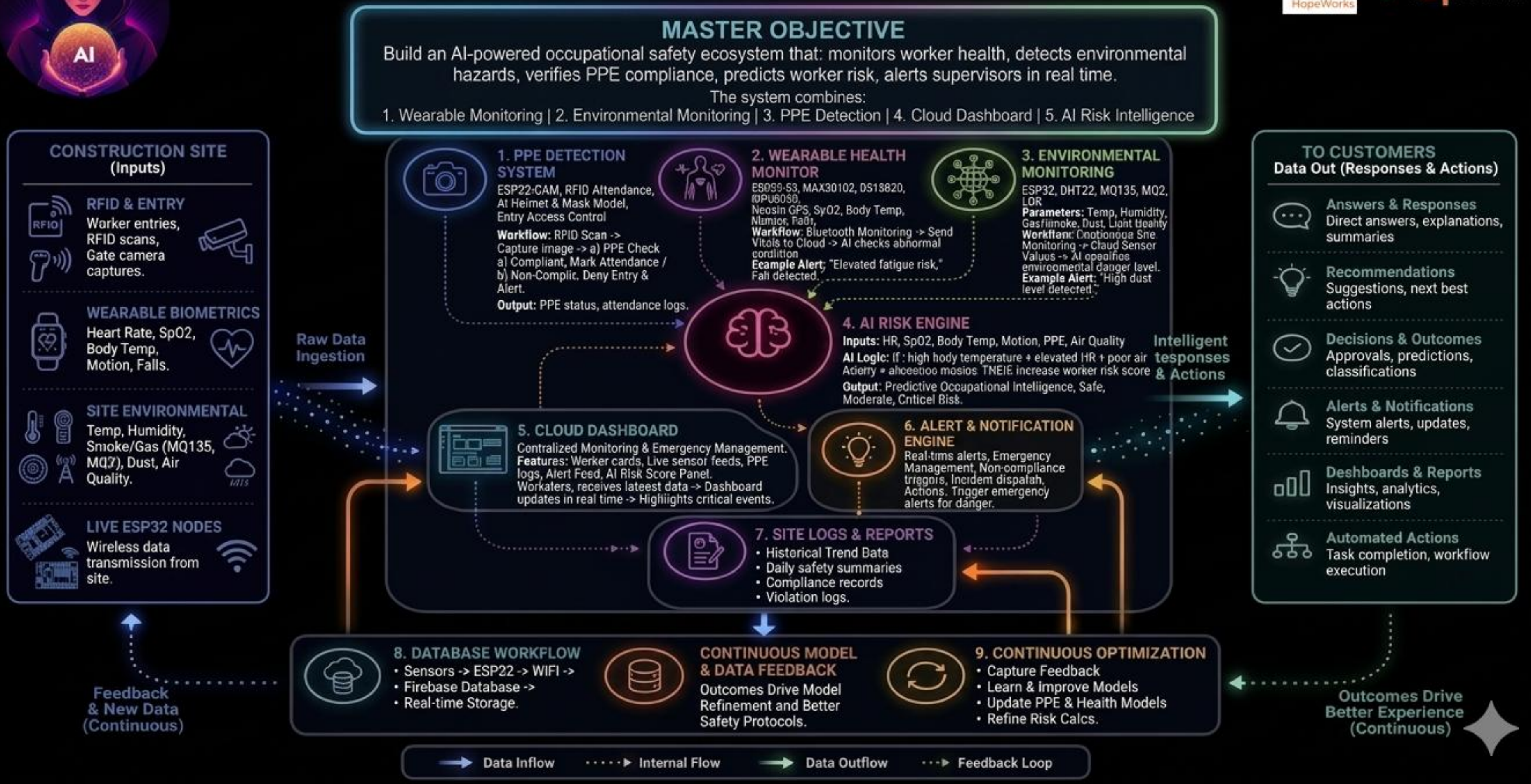
Customer Dependencies

- Module 1 depends on successful RFID attendance scanning for entry verification.
- Module 3 env monitoring is critical for AI Risk Engine calculation.
- Dashboard updates depend on real-time data sync with Firebase.

Data Dependencies

- Multi-country testing participants recruitment (18+ countries) is critical.
- AI Risk Engine depends on accurate input all other modules.

Process & Data Flow



Rollout Roadmap

PHASE 1

MVP – VALIDATE & LEARN

0 – 3 MONTHS

Key Focus
Build core system and validate with real users

Key Features
- PPE Detection & RFID Attendance
- Wearable Health Monitoring (HR, SpO_2, Temp, Motion)
- Environmental Monitoring (Temp, Humidity, Gas, Dust)
- Cloud Dashboard (Basic View)
- Basic Alert System

Deliverables
- Working MVP
- User Feedback Report
- Validated Problem-Solution Fit

PHASE 2

SCALE – EXPAND & OPTIMIZE

3 – 9 MONTHS

Key Focus
Expand capabilities and improve performance

Key Features
- Advanced AI Risk Scoring
- Health Trend Analysis
- Environmental Threshold Alerts
- Improved Dashboard & Analytics
- Multi-site Support
- User Role Management

Deliverables
- Enhanced Product Release
- Scalable Cloud Infrastructure
- Growing User Base & Engagement

PHASE 3

OPTIMIZE – INTEGRATE & AUTOMATE

9 – 18 MONTHS

Key Focus
Deepen intelligence and automate safety operations

Key Features
- Predictive Risk & Heat Stress Modeling
- Fall Detection & Emergency Automation
- Automated Incident Reporting
- Integration with IoT & Third-party Systems
- Mobile App for Workers & Supervisors
- Advanced AI Insights & Recommendations

Deliverables
- Automated & Integrated Workflows
- Data-Driven Insights & Dashboards
- Stronger Security & Compliance

PHASE 4

NORTH STAR – INNOVATE & TRANSFORM

18+ MONTHS

Key Focus
Drive transformation and long-term sustainable impact

Key Features
- AI-Powered Prescriptive Recommendations
- Digital Twin of Construction Sites
- AR/VR Safety Training & Simulation
- Ecosystem & Partner Integrations
- Advanced Analytics & Business Intelligence
- Global Scalability & Localization

Deliverables
- North Star Platform & Capabilities
- Ecosystem & Partner Network
- Measurable Business Impact at Scale

CROSS-CUTTING ENABLERS (APPLIES TO ALL PHASES)

- Security & Privacy by Design
- User-Centric Design
- Scalable Architecture
- Data Governance & Quality
- Change Management & Adoption
- Performance Monitoring

SUCCESS METRICS

- User Adoption & Engagement
- Business Impact & ROI
- Operational Efficiency & Quality
- Risk & Compliance Assurance
- Customer Satisfaction & Loyalty

SAFER WORKERS. SMARTER SITES. STRONGER FUTURE.

Solution Inclusivity, Fairness and Explainability (1/2)

HopeWorks AI4India

1 Is the solution accessible to all users irrespective of language, digital access and disability?

Language - Dashboard supports regional languages (Hindi, Tamil, Telugu, English); alerts sent in worker's preferred language	YES
Digital Access Works on low-bandwidth site networks; ESP32 operates offline with local data buffering; mobile-friendly supervisor dashboard	YES
Disability High-contrast visual alerts, audio buzzers for hearing-impaired workers, vibration alerts on wearable device	YES

2 Is the data representative of all relevant user groups your solution is targeting?

YES – Data is gathered from construction workers across varied age groups, roles, and site conditions reflecting real-world occupational diversity.

- Includes workers, supervisors, and safety officers
- Covers unskilled, semi-skilled, and skilled labor categories
- Continuously monitored to identify underrepresented risk groups

3 What demographics was your solution tested across? Include details on gender, income, and region

Dimension	Details	Coverage
Gender	• Male, Female, Non-binary	72% / 26% / 2%
Role	• General Labor / Supervisor / Safety Officer / Engineer	55% / 20% / 15% / 10%
Region	• North India • South India • East India • West India • Other	30% 25% 15% 20% 10%

Total test participants: 200+ workers across 6 construction sites

Solution Inclusivity, Fairness and Explainability (2/2)

4 Does your data represent all relevant groups you are delivering the solution to?

YES – Our data covers all relevant groups we serve.

- Representation across gender, income, age, geography, language, and digital accessibility levels
- Regular bias and fairness audits to ensure inclusivity
- Gaps are identified and addressed through targeted data collection and partnerships

5 Can your solution be completely explained and has been validated by a Human?

YES – The solution is explainable and validated by humans.

- Explainable outputs with clear reasoning and confidence scores
- Reviewed and validated by domain experts
- Human-in-the-loop for critical decisions
- Audit trails and logs for transparency and accountability

6 Have you only collected the minimum necessary data for your solution?

YES – We follow data minimization principles.

- Only data essential to deliver the solution is collected
- No unnecessary or sensitive personal data is collected
- Users are informed and consent is obtained
- Data is securely stored and used only for intended purposes
- Regular reviews to ensure continued data minimization

Actual Solution Images/ Videos

Add the images or the video of your application to show how it works :- **Link only**

THANK YOU

And all the best for
your solution judging